

## KEY EVENTS

## 11TH–15TH CENTURY

■ **1044** A document written for the Chinese emperor describes the use of gunpowder for smoke bombs and incendiary devices in sieges.

■ **1231** “Fire lances” – primitive flamethrowers – and “thunder-crash bombs” are employed in the defence of Kaifeng (Beijing) against the Mongols’ forces.

■ **1326** The government of Florence, Italy, orders a supply of metal cannon and ammunition to be used in the defence of the city.

■ **1346** King Edward III of England uses small field cannon against the French, at the Battle of Crécy.

■ **c. 1350** Chinese armies are regularly deploying hundreds of cannon and primitive handguns in field battles and sieges.

■ **1380s** Large siege guns (bombards) come into use in Europe; some are capable of firing stone shot weighing up to 230kg (500lb).

■ **1421** Using hand guns and cannon mounted on wagons, Hussites led by Jan Zizka defeat Sigismund, King of Hungary and Bohemia, at the Battle of Kutna Hora.

■ **1439–53** Brothers Jean and Gaspard Bureau create a powerful artillery train for King Charles VII of France, aiding him in the defeat of the English in the Hundred Years War.

■ **1453** Sultan Mehmed II of the Ottomans uses large cannon to decisive effect during the successful siege of Constantinople.

## ► SIMPLE FIREPOWER

Most early gunpowder weapons were crude by later standards, and did not make a great physical impact; their effect on the enemy was mainly psychological.

## KEY DEVELOPMENT

## GUNPOWDER MAKES ITS MARK

The arrival of gunpowder is often dated to 1453, when the massive cannon of Sultan Mehmed II breached the walls of Constantinople, a city that had resisted sieges for a thousand years. However, by this time, gunpowder had already been in military use in China for four centuries.

The explosive properties of a mix of charcoal, sulphur, and saltpetre were probably discovered by Chinese alchemists during the Tang dynasty (618–907CE). Documents from the Song dynasty, in the 11th century, make reference to the use of gunpowder-tipped arrows as incendiary devices, and of gunpowder fire-bombs that were hurled into besieged cities by catapult.

Over the following centuries, a whole family of gunpowder weapons was developed, from flame-throwing “fire-lances” to grenades, rockets, hand-held guns, and primitive cannon. These devices did not transform the Chinese practice of war, but took their place as a significant but subsidiary adjunct to traditional weaponry, often employed primarily for psychological effect.

## THE NEW TECHNOLOGY

As the new gunpowder-based technology filtered through to the Muslim armies of West Asia and to Christian Europe, in the 12th and 13th centuries, its impact at first remained marginal. The few small cannon deployed at Crécy in 1346 were puny in their effect compared with the decisive firepower of the archers’ longbows.

However, European military artisans and their employers saw the potential of large guns for sieges.

And so, by the 15th century, bombards (large siege guns) were being made, usually with barrels of wrought iron rods bound

together with iron hoops. Exploiting the propulsive power of improved “corned” gunpowder, these mighty but cumbersome cannon were capable of hurling stone shot with crushing force against the walls of large structures, such as castles and towns, facilitating early siege warfare. The artillery used at the 1453 Siege of Constantinople, made for Ottoman Sultan Mehmed by a renegade Christian Hungarian artillery expert, included a bombard so large that it travelled on a cart drawn by 50 yoke of oxen and could fire a ball weighing up to half a ton.

For a brief period of time, heavy cannon were decisive weapons. At the same time that Mehmed was taking Constantinople, the French royal artillery, created by brothers Jean and Gaspard Bureau, was making a decisive contribution to the defeat of the English at the end of the Hundred



“We can compose artificially a fire that can be launched over long distances... It is possible with it to destroy a town or an army”

ROGER BACON, *OPUS MAIOR*, 1248